# A SYMBOLIC POWER OF A PRIME IN A NOETHERIAN RING HAS THAT PRIME AS ITS ONLY ASSOCIATED PRIME

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*Allowed standard prerequisites (editorial).*

- Symbolic power as an intersection with a power in the localisation. Used inside the counted proof at line 15789. Rewrites the n-th symbolic power as R intersected with p^n R_p, which is what makes the divisibility test effective.
- Associated primes of a module via its zerodivisors. Used at the end of the counted proof (line 15791). Converts the fact that every element outside p is a nonzerodivisor into the single-element description of the associated primes.

**Lemma 0.1.** *Let $R$ be a Noetherian ring. Let $\mathfrak{p}$ be a prime ideal. Let $n > 0$. Then $Ass(R/\mathfrak{p}^{(n)}) = \{\mathfrak{p}\}$.*

*Proof.* If $\mathfrak{q}$ is an associated prime of $R/\mathfrak{p}^{(n)}$ then clearly $\mathfrak{p} \subset \mathfrak{q}$. On the other hand, any element $x \in R$, $x \notin \mathfrak{p}$ is a nonzerodivisor on $R/\mathfrak{p}^{(n)}$. Namely, if $y \in R$ and $xy \in \mathfrak{p}^{(n)} = R \cap \mathfrak{p}^n R_{\mathfrak{p}}$ then $y \in \mathfrak{p}^n R_{\mathfrak{p}}$, hence $y \in \mathfrak{p}^{(n)}$. Hence the lemma follows. □

*Date*: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.